HARBOR: Heavy-Hitter Aware Eviction for Long-Context KV Caches

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Abstract

Most cached tokens are rarely attended to. HARBOR is a training-free eviction policy that keeps only a fixed budget of the key-value cache: the first tokens of the sequence, a window of recent tokens, and the heavy hitters that have received the most attention so far. At a cache budget of 20% HARBOR keeps 97.8% of the full-cache accuracy on LongSuite.

1 Introduction

Attention in trained language models is sparse: in a typical layer, five percent of the tokens receive more than half of the attention mass. If the tokens that matter can be identified while decoding, the rest of the cache can be dropped without retraining.

2 Method

HARBOR always keeps the first four tokens because they act as attention sinks: the softmax must place its mass somewhere, and models learn to park it on the first positions, so evicting them changes every later attention distribution. It also keeps the 256 most recent tokens, which carry the local context.

The remaining budget goes to heavy hitters. Each cached token keeps a running sum of the attention it has received from all queries so far, and HARBOR ranks tokens by this accumulated attention score. When the cache is full the token with the lowest score outside the sinks and the recent window is evicted.

$$A_j = \sum_t \mathrm{softmax}(q_t K^\top)_j, \quad \text{evict } \arg\min_j A_j \tag{1}$$

3 Results

Table 1: LongSuite accuracy by cache budget, 7B model.

Policy	Budget	LongSuite	Kept
Full cache	100%	47.6	100%
Recent window only	20%	33.0	69.3%
HARBOR	20%	46.6	97.8%
HARBOR + SQUEEZE	20%	46.1	96.8%

Eviction and quantization address different redundancy and combine well: applying HARBOR on top of SQUEEZE two-bit quantization reduces cache memory by 12 times with a loss of 1.5 points.

4 Limitations

An evicted token cannot be recovered. Questions about a detail that received little attention early in a long document are the main failure case, which a quantized cache does not have.

References

[1] P. Raman, O. Weiss and D. Feld. SQUEEZE: Two-bit key-value cache quantization with per-channel keys. arXiv preprint, 2024.

[2] T. Okafor and R. Lind. Attention sinks in long sequences. In Proceedings of the Sequence Modelling Workshop, 2023.